



Shiretechnik Solutions Pvt Ltd

SHape Your Idea To Reality

Case Studies on Wearable Peltier Device

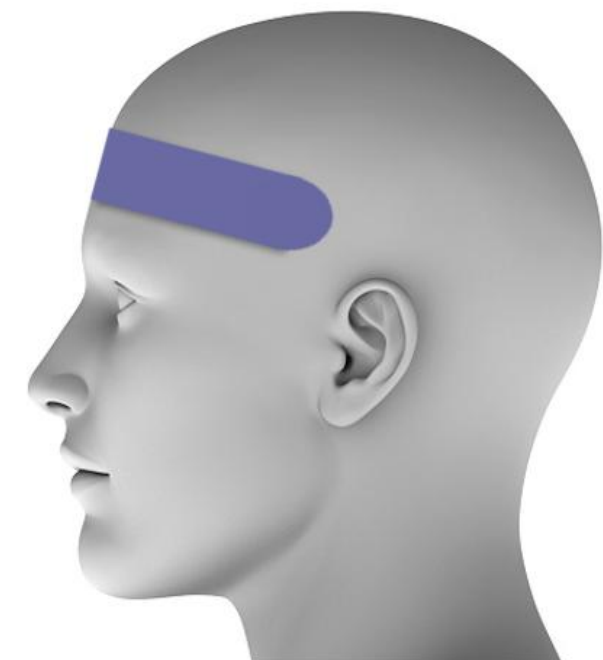
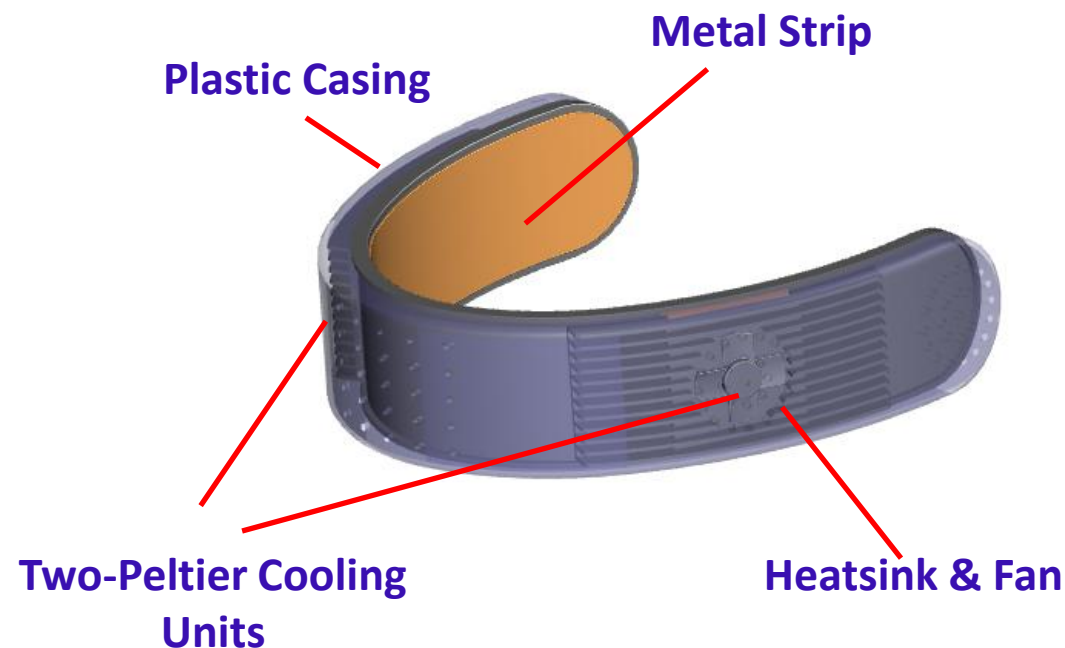


Objective of Thermal Analysis

- To evaluate the thermal performance of Wearable Peltier Device for various use-case conditions pre-defined by the client
- Cooling Design Optimization

Product Overview

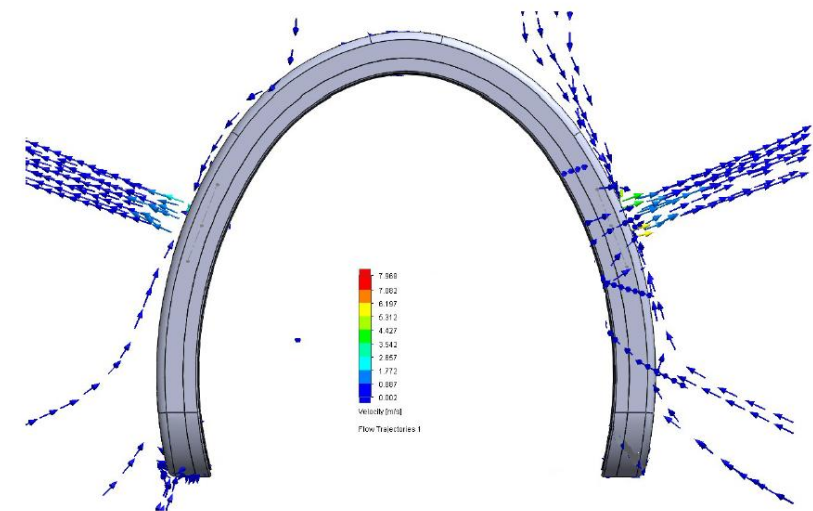
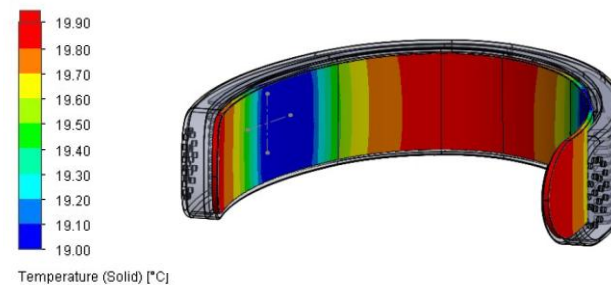
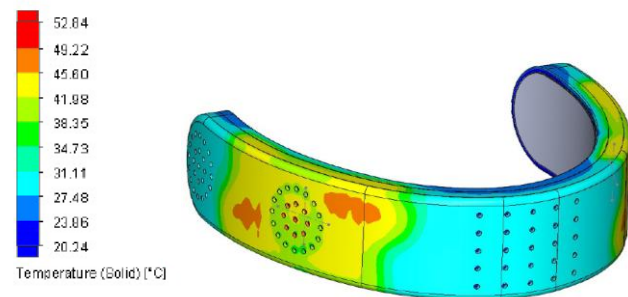
Wearable Peltier Device is a headband version operable to fit around a user's head. The device is capable of cooling/heating the user's head as per the requirement



Key Challenges Faced in Analysis

- A detailed study on heat dissipation from human body for various activities was carried out and an appropriate variable heat load considered in simulation analysis in-line with the real-world applications.
- Most suitable & proven thermal modelling methodology of TEC (Thermo-Electric Cooler) was employed to predict the temperatures accurately.
- Extensive transient studies were also conducted to precisely characterize the device and measured thermal response and time taken by device to reach the required temperatures.
- Most effective heatsink for maximum thermal performance & minimal weight was designed with optimization techniques within the given space constraints.

Analysis Results



- A detailed analysis showed that the thermal performance of device was well within the product's specifications requirement for all the use-cases pre-defined by the client
- Additionally, the optimization analysis was conducted for further refinement of product performance & for a targeted BoM cost

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